

Ryan Regalado

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in Statistics, Minors in Computer Science & Mathematics

Aug. 2024 – Expected: May 2028

- **Achievements:** 3.93 GPA — Dean's List (Fall 2024, Spring 2025)
- **Relevant Coursework:** Mathematical Statistics I & II, Regression Analysis, Statistical Machine Learning, Stochastic Processes, Nonlinear Optimization, Mathematical Optimization for Data Science

EXPERIENCE

Blackwell Research Scholar

Jun. 2026 – Present

University of Illinois Department of Statistics

Champaign, IL

- Developed a statistical simulation framework in Python for analyzing GJIVE, a low-rank data integration model, across high-dimensional datasets and perturbed experimental settings.
- Optimized linear algebra pipelines using IRLB-based truncated SVD calculations and parallel CPU workflows, reducing estimation algorithm runtime by 66%.
- Applied the GJIVE model on fMRI connectivity data (734 subjects, 19K features) to find interpretable low-rank representations associated with autism-related brain regions.
- Delivered various modes of communication: interactive dashboards and simulation plots via Streamlit, detailed LaTeX write-ups, and an organized and well-documented codebase.

Undergraduate Researcher

Jan. 2026 – May 2026

Illinois Mathematics Lab

Champaign, IL

- Analyzed end-of-season collapse patterns in win/loss sequences across the NBA, MLB, and European soccer leagues by creating probabilistic metrics to identify major historical season collapses.
- Co-developed a data engineering pipeline and database schema integrating 500,000+ games from multiple sources.
- Organized a collaborative repository of 300+ notebooks, scripts, datasets, and figures across the research team.
- Accelerated simulation-heavy outputs by 18× through vectorized computations.

PROJECTS

Hotel Price Prediction | *R, tidyverse, caret, glmnet, randomForest*

Apr. 2026 – May 2026

- Developed and evaluated eight regression and classification models—including linear regression, LASSO, KNN, bagging, random forest, LDA, QDA, and Naive Bayes—on 110,000+ hotel bookings with 31 predictors.
- Achieved a best test RMSE of 24.9 using Random Forest and 66.3% classification accuracy using LDA through comparative model evaluation.
- Engineered quantile-based ADR classes and applied PCA to mitigate multicollinearity for QDA and Naive Bayes models.

College Loan Visualization | *JavaScript, React, Express.js, Node.js, Supabase*

Jun. 2025 – Sep. 2025

- Developed a full-stack loan analysis tool using Node.js, React, Express.js, and Supabase, leveraging Supabase API for anonymous user authentication—securely tracking user-specific data for personalized financial profiles.
- Designed databases, routing, and frontend behavior while leveraging AI for rapid development of CSS styling.
- Implemented logic for comprehensive repayment & net income calculations using state-specific tax information.

Fraud Call Detection | *Python, pandas, scikit-learn, NLTK, TextBlob*

Jan. 2025 – May 2025

- Built an NLP-based fraud call classifier to help identify scam voicemail transcripts for consumer protection.
- Developed NLP pipeline using Regex and TextBlob, preparing a corpus of 5000+ call transcripts for vectorization via Term Frequency-Inverse Document Frequency (TF-IDF).
- Trained a logistic regression model for fraud detection, accounting for a 9:1 class imbalance ratio, yielding 81% precision.
- Deployed classification model on iPhone voicemail transcripts, achieving 80% accuracy on real-world test data.

TECHNICAL SKILLS

Languages: Python, R, JavaScript, Java, C++

Frameworks & Platforms: React, Node.js, Express.js, Supabase, Streamlit

Libraries: pandas, NumPy, SciPy, scikit-learn, Matplotlib, NLTK, TextBlob | tidyverse, ggplot2, glmnet, randomForest

Developer Tools: Git, LaTeX, VS Code